

# Database diagram and data dictionary

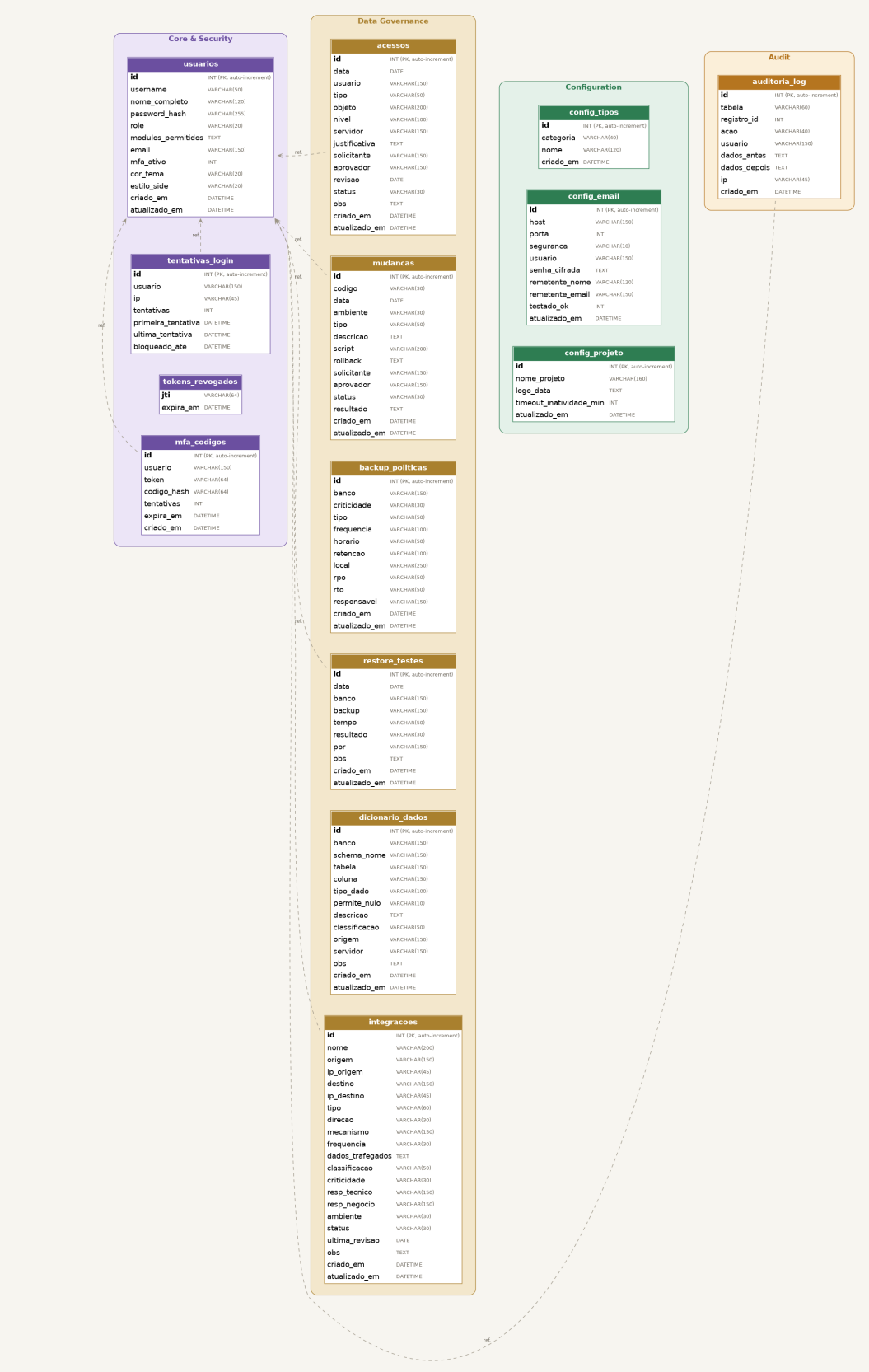
Internal database of the Data Management Portal

Note: table and column names are in Portuguese (the actual database identifiers). All descriptions are in English.

Generated on 2026-07-02

# Database diagram

Each box represents a table; the bold column is the primary key. Dashed lines indicate logical references by text value (there are no real foreign-key constraints in the database — for example, *integracoes.resp\_tecnico* stores the same value as *usuarios.username*).



# Data dictionary

Complete list of the portal's internal database tables and columns, organised by the same groups shown in the diagram. Column names are the real Portuguese identifiers used in the database.

## Core & Security

### usuarios

| Column             | Type                     | Constraints                 | Description (EN)                                   |
|--------------------|--------------------------|-----------------------------|----------------------------------------------------|
| id                 | INT (PK, auto-increment) |                             | Internal user identifier.                          |
| username           | VARCHAR(50)              | NOT NULL, UNIQUE            | Login used for authentication.                     |
| nome_completo      | VARCHAR(120)             |                             | Full name displayed in the interface.              |
| password_hash      | VARCHAR(255)             | NOT NULL                    | Password hash (never stored as plain text).        |
| role               | VARCHAR(20)              | NOT NULL, default 'leitura' | Access role (e.g. admin, leitura/read).            |
| modulos_permitidos | TEXT                     |                             | Serialised list of modules allowed for this user.  |
| email              | VARCHAR(150)             |                             | E-mail used for notifications and access recovery. |
| mfa_ativo          | INT                      | NOT NULL, default 0         | 1 when two-step authentication is active.          |
| cor_tema           | VARCHAR(20)              | NOT NULL, default 'padrao'  | Colour theme chosen by the user.                   |
| estilo_side        | VARCHAR(20)              | NOT NULL, default 'claro'   | Sidebar style chosen by the user.                  |
| criado_em          | DATETIME                 |                             | Record creation timestamp.                         |
| atualizado_em      | DATETIME                 |                             | Last update timestamp.                             |

### tentativas\_login

| Column             | Type                     | Constraints         | Description (EN)                                |
|--------------------|--------------------------|---------------------|-------------------------------------------------|
| id                 | INT (PK, auto-increment) |                     | Internal identifier.                            |
| usuario            | VARCHAR(150)             | NOT NULL            | Login that attempted to authenticate.           |
| ip                 | VARCHAR(45)              | NOT NULL            | Source IP address of the attempt.               |
| tentativas         | INT                      | NOT NULL, default 0 | Counter of consecutive failed attempts.         |
| primeira_tentativa | DATETIME                 |                     | When the sequence of attempts started.          |
| ultima_tentativa   | DATETIME                 |                     | When the most recent attempt occurred.          |
| bloqueado_ate      | DATETIME                 |                     | Until when the login is locked (if applicable). |

### tokens\_revogados

| Column    | Type        | Constraints | Description (EN)                                   |
|-----------|-------------|-------------|----------------------------------------------------|
| jti       | VARCHAR(64) | PK          | Unique identifier of the revoked JWT token.        |
| expira_em | DATETIME    |             | Original token expiry; used for automatic cleanup. |

### mfa\_codigos

| Column      | Type                     | Constraints         | Description (EN)                                       |
|-------------|--------------------------|---------------------|--------------------------------------------------------|
| id          | INT (PK, auto-increment) |                     | Internal identifier.                                   |
| usuario     | VARCHAR(150)             | NOT NULL            | Login associated with the verification code.           |
| token       | VARCHAR(64)              | NOT NULL            | Identifies the pending login attempt on the front-end. |
| codigo_hash | VARCHAR(64)              | NOT NULL            | HMAC hash of the 6-digit code sent by e-mail.          |
| tentativas  | INT                      | NOT NULL, default 0 | Counter of wrong-code attempts.                        |
| expira_em   | DATETIME                 |                     | Verification code expiry.                              |
| criado_em   | DATETIME                 |                     | Code generation timestamp.                             |

## Data Governance

### acessos

| Column        | Type                     | Constraints     | Description (EN)                                          |
|---------------|--------------------------|-----------------|-----------------------------------------------------------|
| id            | INT (PK, auto-increment) |                 | Internal identifier.                                      |
| data          | DATE                     |                 | Date access was granted/reviewed.                         |
| usuario       | VARCHAR(150)             |                 | User or account that received access (free text).         |
| tipo          | VARCHAR(50)              |                 | Type of access granted.                                   |
| objeto        | VARCHAR(200)             |                 | Object/resource to which access refers.                   |
| nivel         | VARCHAR(100)             |                 | Privilege level granted.                                  |
| servidor      | VARCHAR(150)             |                 | Server/instance where access was granted.                 |
| justificativa | TEXT                     |                 | Reason for granting access.                               |
| solicitante   | VARCHAR(150)             |                 | Who requested the access (text ref to usuarios.username). |
| aprovador     | VARCHAR(150)             |                 | Who approved the access (text ref to usuarios.username).  |
| revisao       | DATE                     |                 | Scheduled date for periodic access review.                |
| status        | VARCHAR(30)              | default 'Ativo' | Current access status.                                    |
| obs           | TEXT                     |                 | Free-form notes.                                          |
| criado_em     | DATETIME                 |                 | Record creation timestamp.                                |
| atualizado_em | DATETIME                 |                 | Last update timestamp.                                    |

### mudancas

| Column    | Type                     | Constraints | Description (EN)                                 |
|-----------|--------------------------|-------------|--------------------------------------------------|
| id        | INT (PK, auto-increment) |             | Internal identifier.                             |
| codigo    | VARCHAR(30)              |             | Change control code (RFC/ticket).                |
| data      | DATE                     |             | Change execution date.                           |
| ambiente  | VARCHAR(30)              |             | Affected environment (e.g. production, staging). |
| tipo      | VARCHAR(50)              |             | Type of change.                                  |
| descricao | TEXT                     |             | Description of what was changed.                 |

| Column        | Type         | Constraints | Description (EN)                                          |
|---------------|--------------|-------------|-----------------------------------------------------------|
| script        | VARCHAR(200) |             | Name/reference of the script executed.                    |
| rollback      | TEXT         |             | Rollback procedure, if needed.                            |
| solicitante   | VARCHAR(150) |             | Who requested the change (text ref to usuarios.username). |
| aprovador     | VARCHAR(150) |             | Who approved the change (text ref to usuarios.username).  |
| status        | VARCHAR(30)  |             | Change status.                                            |
| resultado     | TEXT         |             | Execution result.                                         |
| criado_em     | DATETIME     |             | Record creation timestamp.                                |
| atualizado_em | DATETIME     |             | Last update timestamp.                                    |

### backup\_politicas

| Column        | Type                     | Constraints | Description (EN)                       |
|---------------|--------------------------|-------------|----------------------------------------|
| id            | INT (PK, auto-increment) |             | Internal identifier.                   |
| banco         | VARCHAR(150)             |             | Database covered by this policy.       |
| criticidade   | VARCHAR(30)              |             | Database criticality.                  |
| tipo          | VARCHAR(50)              |             | Backup type (full, incremental, etc.). |
| frequencia    | VARCHAR(100)             |             | Execution frequency.                   |
| horario       | VARCHAR(50)              |             | Scheduled time.                        |
| retencao      | VARCHAR(100)             |             | Backup retention period.               |
| local         | VARCHAR(250)             |             | Backup storage location.               |
| rpo           | VARCHAR(50)              |             | Recovery Point Objective.              |
| rto           | VARCHAR(50)              |             | Recovery Time Objective.               |
| responsavel   | VARCHAR(150)             |             | Policy owner.                          |
| criado_em     | DATETIME                 |             | Record creation timestamp.             |
| atualizado_em | DATETIME                 |             | Last update timestamp.                 |

### restore\_testes

| Column        | Type                     | Constraints | Description (EN)                                        |
|---------------|--------------------------|-------------|---------------------------------------------------------|
| id            | INT (PK, auto-increment) |             | Internal identifier.                                    |
| data          | DATE                     |             | Restore test date.                                      |
| banco         | VARCHAR(150)             |             | Database tested.                                        |
| backup        | VARCHAR(150)             |             | Backup used in the test.                                |
| tempo         | VARCHAR(50)              |             | Total restore duration.                                 |
| resultado     | VARCHAR(30)              |             | Test result: success/failure.                           |
| por           | VARCHAR(150)             |             | Who performed the test (text ref to usuarios.username). |
| obs           | TEXT                     |             | Free-form notes.                                        |
| criado_em     | DATETIME                 |             | Record creation timestamp.                              |
| atualizado_em | DATETIME                 |             | Last update timestamp.                                  |

### dicionario\_dados

| Column        | Type                     | Constraints | Description (EN)                                 |
|---------------|--------------------------|-------------|--------------------------------------------------|
| id            | INT (PK, auto-increment) |             | Internal identifier.                             |
| banco         | VARCHAR(150)             |             | External database being catalogued.              |
| schema_nome   | VARCHAR(150)             |             | Schema of the catalogued table.                  |
| tabela        | VARCHAR(150)             |             | Name of the catalogued table.                    |
| coluna        | VARCHAR(150)             |             | Name of the catalogued column.                   |
| tipo_dado     | VARCHAR(100)             |             | Data type of the catalogued column.              |
| permite_nulo  | VARCHAR(10)              |             | Indicates whether the column accepts nulls.      |
| descricao     | TEXT                     |             | Description/meaning of the column.               |
| classificacao | VARCHAR(50)              |             | Data sensitivity classification.                 |
| origem        | VARCHAR(150)             |             | Data origin system/process.                      |
| servidor      | VARCHAR(150)             |             | Server/instance hosting the catalogued database. |
| obs           | TEXT                     |             | Free-form notes.                                 |
| criado_em     | DATETIME                 |             | Record creation timestamp.                       |
| atualizado_em | DATETIME                 |             | Last update timestamp.                           |

## integracoes

| Column           | Type                     | Constraints | Description (EN)                                 |
|------------------|--------------------------|-------------|--------------------------------------------------|
| id               | INT (PK, auto-increment) |             | Internal identifier.                             |
| nome             | VARCHAR(200)             |             | Integration name.                                |
| origem           | VARCHAR(150)             |             | Source system.                                   |
| ip_origem        | VARCHAR(45)              |             | IP of the source system.                         |
| destino          | VARCHAR(150)             |             | Target system.                                   |
| ip_destino       | VARCHAR(45)              |             | IP of the target system.                         |
| tipo             | VARCHAR(60)              |             | Integration type.                                |
| direcao          | VARCHAR(30)              |             | Data flow direction.                             |
| mecanismo        | VARCHAR(150)             |             | Integration mechanism/protocol.                  |
| frequencia       | VARCHAR(30)              |             | Execution frequency.                             |
| dados_trafegados | TEXT                     |             | Description of the data transmitted.             |
| classificacao    | VARCHAR(50)              |             | Data sensitivity classification.                 |
| criticidade      | VARCHAR(30)              |             | Integration criticality.                         |
| resp_tecnico     | VARCHAR(150)             |             | Technical owner (text ref to usuarios.username). |
| resp_negocio     | VARCHAR(150)             |             | Business owner (text ref to usuarios.username).  |
| ambiente         | VARCHAR(30)              |             | Integration environment.                         |
| status           | VARCHAR(30)              |             | Current integration status.                      |
| ultima_revisao   | DATE                     |             | Date of last review.                             |
| obs              | TEXT                     |             | Free-form notes.                                 |
| criado_em        | DATETIME                 |             | Record creation timestamp.                       |
| atualizado_em    | DATETIME                 |             | Last update timestamp.                           |

## Configuration

### config\_tipos

| Column    | Type                     | Constraints | Description (EN)                                  |
|-----------|--------------------------|-------------|---------------------------------------------------|
| id        | INT (PK, auto-increment) |             | Internal identifier.                              |
| categoria | VARCHAR(40)              | NOT NULL    | List discriminator (e.g. ambiente, tipo_mudanca). |
| nome      | VARCHAR(120)             | NOT NULL    | Value/option registered for the category.         |
| criado_em | DATETIME                 |             | Record creation timestamp.                        |

### config\_email

| Column          | Type                     | Constraints         | Description (EN)                                      |
|-----------------|--------------------------|---------------------|-------------------------------------------------------|
| id              | INT (PK, auto-increment) |                     | Single row (id = 1).                                  |
| host            | VARCHAR(150)             |                     | SMTP server hostname.                                 |
| porta           | INT                      |                     | SMTP port.                                            |
| seguranca       | VARCHAR(10)              | default 'tls'       | SMTP connection security type.                        |
| usuario         | VARCHAR(150)             |                     | SMTP authentication username.                         |
| senha_cifrada   | TEXT                     |                     | Encrypted SMTP password (never stored as plain text). |
| remetente_nome  | VARCHAR(120)             |                     | Name displayed as the sender.                         |
| remetente_email | VARCHAR(150)             |                     | E-mail address used as sender.                        |
| testado_ok      | INT                      | NOT NULL, default 0 | 1 when a test send succeeded.                         |
| atualizado_em   | DATETIME                 |                     | Last update timestamp.                                |

### config\_projeto

| Column                  | Type                     | Constraints          | Description (EN)                            |
|-------------------------|--------------------------|----------------------|---------------------------------------------|
| id                      | INT (PK, auto-increment) |                      | Single row (id = 1).                        |
| nome_projeto            | VARCHAR(160)             |                      | Customised portal name.                     |
| logo_data               | TEXT                     |                      | Customised logo as a data URI (base64).     |
| timeout_inatividade_min | INT                      | NOT NULL, default 30 | Inactivity minutes before automatic logout. |
| atualizado_em           | DATETIME                 |                      | Last update timestamp.                      |

## Audit

### auditoria\_log

| Column      | Type                     | Constraints | Description (EN)                 |
|-------------|--------------------------|-------------|----------------------------------|
| id          | INT (PK, auto-increment) |             | Internal identifier.             |
| tabela      | VARCHAR(60)              | NOT NULL    | Table affected by the operation. |
| registro_id | INT                      |             | ID of the affected record.       |

| Column       | Type         | Constraints | Description (EN)                                           |
|--------------|--------------|-------------|------------------------------------------------------------|
| acao         | VARCHAR(40)  | NOT NULL    | Action performed (criar, editar, excluir, login, etc.).    |
| usuario      | VARCHAR(150) |             | Who performed the action (text ref to usuarios.username).  |
| dados_antes  | TEXT         |             | Record state before the operation (JSON), when applicable. |
| dados_depois | TEXT         |             | Record state after the operation (JSON), when applicable.  |
| ip           | VARCHAR(45)  |             | Source IP of the action.                                   |
| criado_em    | DATETIME     |             | Event timestamp.                                           |